

Design Project 5

Power Flow Analysis: Shunt & Series Compensation in a 10-Bus Power System

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1. **System Overview**
2. **Problem Statement**
3. **Modeling & Setup**
4. **Results**
5. **Conclusion & Recommendations**

Transmission

7 x 230-kV Lines

Voltage Levels

Transformers

4 transformers

Generators

Swing Bus

Bus 1 - $V = 1.0 \angle 0$



Shunt Compensation

Goal: We want to keep all voltages between

0.99 and 1.02 p.u

Two separate settings

- Heavy load
- Light load (25%)

Series Compensation

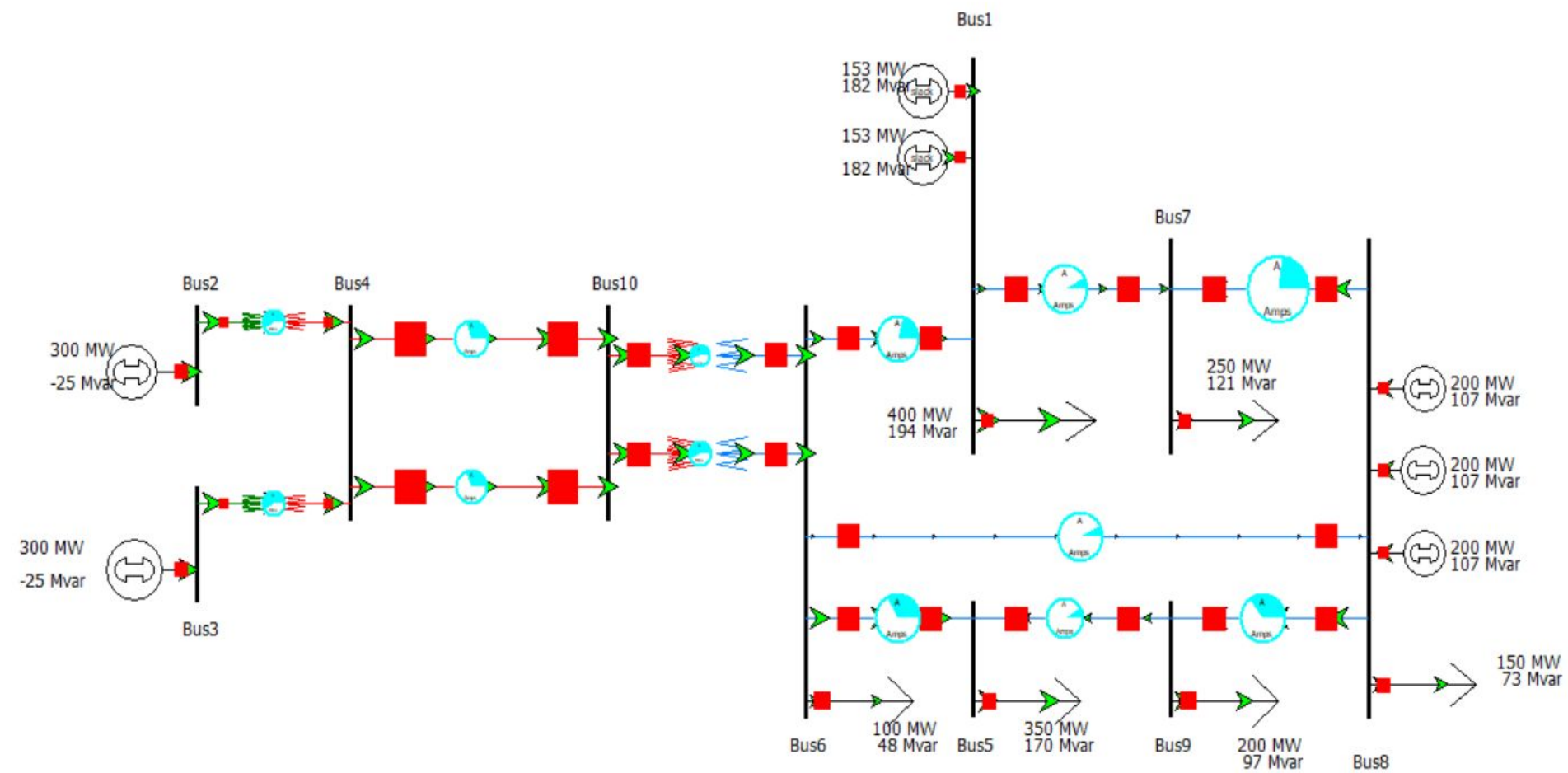
Goal: Limit Bus 4 and 10 angle under one 345-kV transmission line shut down

$$\delta \leq 40$$

- Heavy load

Bus Data Summary

Bus	kV Level	Type	V (pu)	P Gen (pu)	P Load (pu)	QLoad (pu)	Q Limit (pu)
1	230	Swing	1.0 $\angle$ 0	-	4.0	1.94	-
2	22	PV	1.0	3.0	-	-	-1.45 to +1.45
3	22	PV	1.0	3.0	-	-	-1.45 to +1.45
4	345	PQ	-	-	-	-	-
5	230	PQ	-	-	3.5	1.7	-
6	230	PQ	-	-	1.0	0.48	-
7	230	PQ	-	-	2.5	1.21	-
8	230	PV	1.0	6.0	1.5	0.726	-1.95 to +1.95
9	230	PQ	-	-	2.0	0.97	-
10	345	PQ	-	-	-	-	-



X Buses X Branches Input X Generators X Loads										
Filter Advanced Bus Find... Remove Quick Filter										
	Number	Name	Area Name	Nom kV	PU Volt	Volt (kV)	Angle (Deg)	Gen MW	Gen Mvar	L
1	1	Bus1	1	230.00	1.00000	230.000	0.00	306.19	363.94	
2	2	Bus2	1	22.00	1.00725	22.160	51.85	300.00	-25.38	
3	3	Bus3	1	22.00	1.00725	22.160	51.85	300.00	-25.38	
4	4	Bus4	1	345.00	0.96869	334.198	48.14			
5	5	Bus5	1	230.00	0.81868	188.297	-30.01			
6	6	Bus6	1	230.00	0.88299	203.089	17.77			
7	7	Bus7	1	230.00	0.85134	195.808	-6.48			
8	8	Bus8	1	230.00	1.01052	232.421	13.78	600.00	320.34	
9	9	Bus9	1	230.00	0.80695	185.598	-17.72			
10	10	Bus10	1	345.00	0.88327	304.727	20.84			

	Number	Name	Area Name	Nom kV	PU Volt	Volt (kV)	Angle (Deg)	Gen MW	Gen Mvar	Load MW	Load Mvar	Dist MW	Dist Mvar	Switched Shunts Mvar
1	1	Bus1	1	230.00	1.00000	230.000	0.00	305.78	257.61	400.00	194.00	0.00	0.00	
2	2	Bus2	1	22.00	1.04325	22.951	45.29	300.00	22.82					
3	3	Bus3	1	22.00	1.04325	22.951	45.29	300.00	22.82					
4	4	Bus4	1	345.00	1.04036	358.926	42.01							
5	5	Bus5	1	230.00	0.81337	187.076	-30.78			350.00	170.00	0.00	0.00	
6	6	Bus6	1	230.00	0.94747	217.918	15.60			100.00	48.00	0.00	0.00	
7	7	Bus7	1	230.00	0.96292	221.472	-6.56			250.00	1.21	0.00	0.00	
8	8	Bus8	1	230.00	1.01003	232.306	12.29	600.00	232.73	150.00	72.60	0.00	0.00	
9	9	Bus9	1	230.00	0.79844	183.642	-19.01			200.00	97.00	0.00	0.00	
10	10	Bus10	1	345.00	0.95413	329.176	18.30							

Heavy Load

Filter	Advanced	Bus					Find...	Remove	Quick
	Number	Name	Area Name	Nom kV	PU Volt	Volt (kV)	Angle (Deg)	Ge	
1	1	Bus1	1	230.00	1.00000	230.000	0.00		
2	2	Bus2	1	22.00	1.00730	22.161	50.43		
3	3	Bus3	1	22.00	1.00730	22.161	50.43		
4	4	Bus4	1	345.00	0.97063	334.866	46.71		
5	5	Bus5	1	230.00	0.89700	206.311	-23.73		
6	6	Bus6	1	230.00	0.91687	210.879	17.45		
7	7	Bus7	1	230.00	0.84278	193.840	-6.28		
8	8	Bus8	1	230.00	1.00884	232.034	14.35		
9	9	Bus9	1	230.00	0.88724	204.065	-14.07		
10	10	Bus10	1	345.00	0.91597	316.010	20.31		

We believed heavy load needed capacitive shunt as lines consume reactive power (I^2X) under heavy current. We wanted to inject reactive power to raise voltages.

It did end up raising some of the buses closer to the target voltage but still fell short.

Light Load

Filter	Advanced	Bus					Find...	Remove	Quit
	Number	Name	Area Name	Nom kV	PU Volt	Volt (kV)	Angle (Deg)		
1	1	Bus1	1	230.00	1.00000	230.000	0.00		
2	2	Bus2	1	22.00	1.00000	22.000	15.73		
3	3	Bus3	1	22.00	1.00000	22.000	15.73		
4	4	Bus4	1	345.00	0.95798	330.503	14.74		
5	5	Bus5	1	230.00	0.00000	0.000	113.20		
6	6	Bus6	1	230.00	0.84854	195.164	7.21		
7	7	Bus7	1	230.00	0.91875	211.313	-1.24		
8	8	Bus8	1	230.00	0.87558	201.383	4.62		
9	9	Bus9	1	230.00	0.44704	102.820	-2.63		
10	10	Bus10	1	345.00	0.86731	299.222	8.07		

We believed light load needed inductive shunt because line charging generated reactive power when lightly loaded, so voltages rise. Reactors absorb excessive power.

Again we got closer to our target voltage values, but fell short.

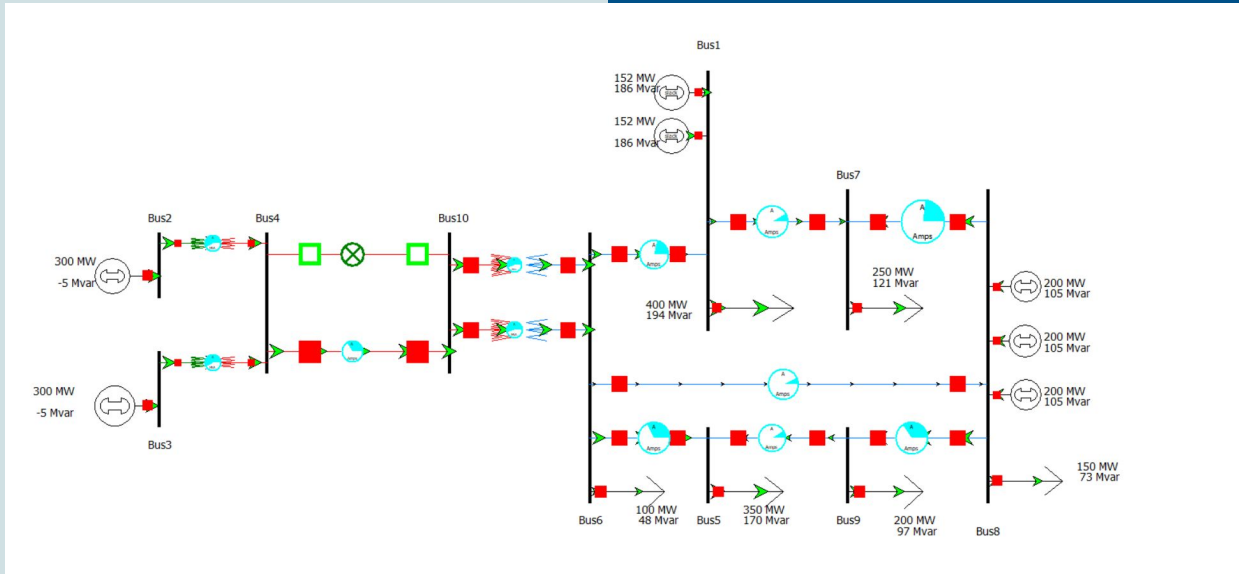
Series Compensation

Our Angle Constraint Was met.

Bus 4 angle: 52.75

Bus 10 Angle: 21.10

Filter		Advanced		Bus		Find...		Remove		Quick	
	Number	Name	Area Name	Nom kV	PU Volt	Volt (kV)	Angle (Deg)	G			
1	1	Bus1	1	230.00	1.00000	230.000	0.00				
2	2	Bus2	1	22.00	1.00992	22.218	56.47				
3	3	Bus3	1	22.00	1.00992	22.218	56.47				
4	4	Bus4	1	345.00	0.96682	333.554	52.75				
5	5	Bus5	1	230.00	0.83689	192.486	-28.50				
6	6	Bus6	1	230.00	0.87568	201.407	18.01				
7	7	Bus7	1	230.00	0.84677	194.757	-6.28				
8	8	Bus8	1	230.00	1.01089	232.505	14.24				
9	9	Bus9	1	230.00	0.82101	188.832	-16.71				
10	10	Bus10	1	345.00	0.87435	301.649	21.10				



1. Heavy Load Shunt Compensation

Capacitive compensation improved voltages but full convergence was not achieved

2. Light Load Shunt Compensation

Inductive compensation was applied but convergence also not achieved. Potential Modeling error with Bus 5 output 0.000 pu.

3. Series Compensation

Bus 4 and 10 Angle difference: 31.65 degrees

Satisfies constraint

Key Takeaways

1.